



PREDICTIVE ANALYTICS FOR GAME-THEORETIC MARKET STRATEGY OPTIMIZATION IN A CLOSED ECONOMY

Hikmet Topak
hikmettopak@marun.edu.tr

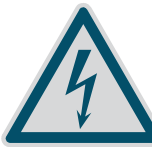
Efe Yalım
efeyalim@marun.edu.tr

Emirhan Ünsal
emirhanunsal@marun.edu.tr

Advisors: Dr. Kamil Erdayandı - Dr. Mehmet Kadir Baran



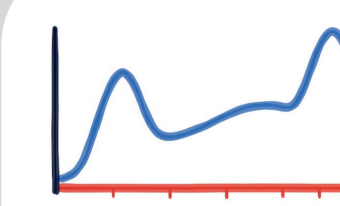
Introduction & Motivation



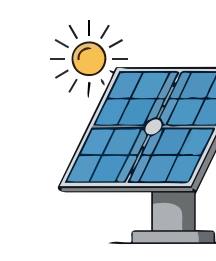
- The Context:** Distributed renewables turn consumers into prosumers, driving a shift to decentralized P2P trading.
- The Problem:** Static, rule-based pricing cannot adapt to real-time supply/demand, causing market inefficiencies and energy waste.
- The Solution:** An automated P2P trading simulation where agents learn dynamic pricing via Independent Multi-Agent Tabular Q-Learning to achieve mutually beneficial pricing outcomes for both buyers and sellers.

System Architecture & Methodology

- Traditional Grid:** Regulated by central utilities. Prosumers suffer from low **Feed-in Tariffs (FiT)**, while consumers face expensive **Time-of-Use (ToU)** rates. Centralized data collection also creates privacy risks.
- P2P Market Shift:** A decentralized approach enabling direct local trading. It protects user privacy, maximizes clearing efficiency, and offers mutually beneficial prices



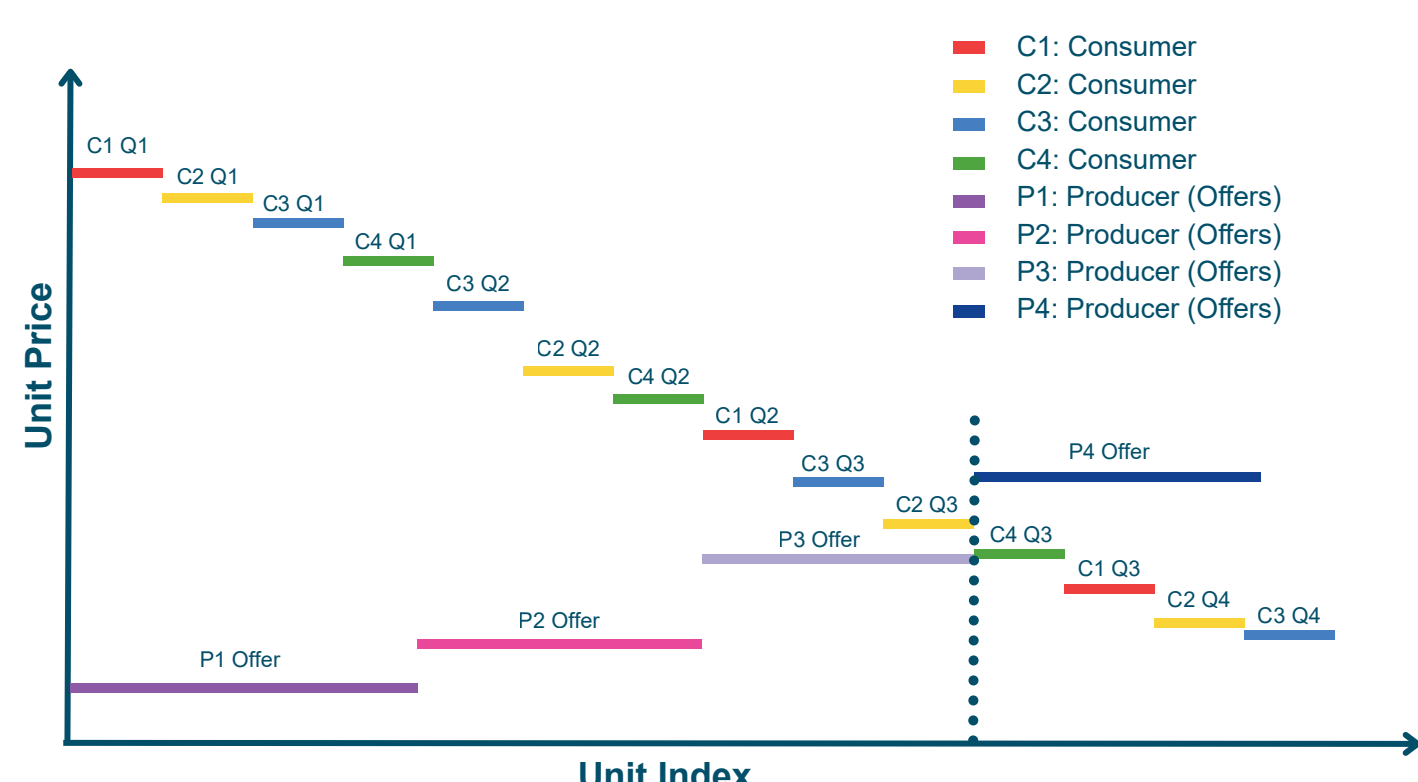
Load profile from Ausgrid dataset (synthetic): peaks at 08:00 & 18:30



PV generation modeled as Gaussian curve (06:00-19:00)

Auction Mechanisms

- a) Continuous Double Auction:** A real-time matching mechanism where transactions occur continuously as soon as individual bids and asks match. (Explored initially but not selected).
- b) Discrete-Time Double Auction (Selected Approach):** Based on the mechanism proposed in the literature, we selected a discrete-time auction that aggregates orders periodically to maximize social welfare.



Fundamentals of the Baseline Model

Market Environment

- Structure: Decentralized P2P electricity market with autonomous producer and consumer agents.
- Commodity: Electrical energy is homogeneous and non-storable. Unmatched sellers export surplus to the grid at the Feed-in Tariff (FiT) rate; unmatched buyers import from the grid at the Time-of-Use (ToU) rate.
- Mechanism: Double Auction under partial information, enabling adaptive learning and price discovery.

Agent Pricing Functions

- Buyer Valuation:** Consumers model their willingness-to-pay via an exponential Marginal Benefit (MB) function, where demand decreases with quantity:

$$MB(q, t) = \alpha \times e^{-\beta q}$$

- Seller Pricing - Use It or Lose It:** Due to near-zero marginal production costs, sellers submit asks via a UILLI pricing function. Higher surplus quantities drive lower ask prices, incentivizing trade over grid export at the minimum FiT rate:

$$P_{UILLI}(q) = FiT + (ToU - FiT) \times e^{-\lambda q}$$

- This baseline market model serves as the benchmark environment for evaluating and comparing the performance of Q-Learning and Neural Network-based pricing agents.

Agent Structure

- 4 Producer (Prosumer) Agents
- 4 Consumer Agents
- 48 time steps per day (30-min intervals)
- Net position at each step:

$$P_{net}(t) = P_{load}(t) - P_{pv}(t)$$

$$P_{net} > 0 \rightarrow \text{Buyer}$$

$$P_{net} < 0 \rightarrow \text{Seller}$$

Key Assumptions

- No energy storage
- No transaction costs
- Incomplete information
- Heterogeneous agents
- Use-It-or-Lose-It (UILLI) seller pricing
- The grid buys surplus energy at low FiT rates and sells it back at higher ToU prices.

Experiments with Q-Learning

State Space

Each agent can simultaneously act as both a seller and a buyer. The role is determined according to the net load (P_{net})

Role	State	Size
Seller	time_slot, PV_surplus_level	3 x 4 = 12 States
Buyer	time_slot, demand_level	3 x 4 = 12 States

PV Surplus / Demand Levels (4 bucket - kWh):

- Level 0: [0,1) kWh - low
- Level 1: [1,3) kWh - medium-low
- Level 2: [3,6) kWh - medium-high
- Level 3: [6,∞) kWh - high

Time slots (3 buckets based on ToU periods):

- 0) Night/Off-peak 00.00-08.00 and 20.00-24.00
- 1) Daytime/Solar 08.00-17.00
- 2) Evening Peak 17.00-20.00

Reward Function

For seller:

$$r_{seller} = q_{matched} \times p_{market} + q_{unmatched} \times FiT + \beta(q_{matched} \times p_{market} - q_{matched} \times FiT)$$
$$\beta = 1.5 \text{ (price premium coefficient signals the agent to "sell above FiT")}$$

Unmatched energy is exported to the grid at FiT rate.

For Buyer $r_{buyer} = -(q_{bought} \times p_{market} + q_{unmet} \times ToU)$

The buyer minimizes cost; unmet demand is penalized at the ToU price.

Q-Learning Update Rule

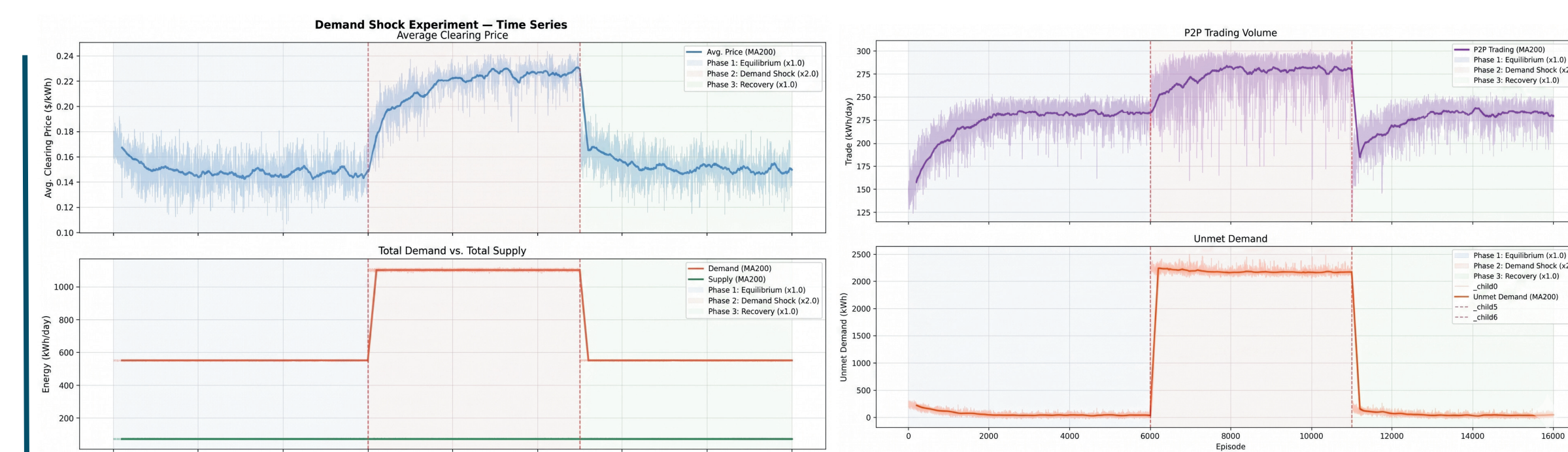
For seller:

$$Q((s_h, s_p), a) \leftarrow Q((s_h, s_p), a) + \alpha [r + \gamma \max_{a'} Q((s'_h, s'_p), a') - Q((s_h, s_p), a)]$$

For buyer:

$$Q((s_h, s_d), a) \leftarrow Q((s_h, s_d), a) + \alpha [r + \gamma \max_{a'} Q((s'_h, s'_d), a') - Q((s_h, s_d), a)]$$

Hyperparameter	Symbol	Value	Description
Learning Rate	α	0.1	How much the Q-value is updated at each step
Discount Factor	γ	0.95	The present value of future rewards (high = far-sighted)
Initial Epsilon	ϵ_0	1	Completely random exploration (episode 0)
Minimum Epsilon	ϵ_{min}	0.05	A constant 5% exploration margin is always maintained
Epsilon Decay	ϵ_{decay}	0.985	At the end of each episode, ϵ is multiplied by this value



Advantages

- Transparent & Interpretable:** Explicit Q-values make the learned policy fully auditable and inspectable.
- Computationally Lightweight:** Two compact 180-cell Q-tables converge reliably without specialized hardware.
- Theoretical Guarantee:** Tabular Q-Learning mathematically guarantees convergence to the optimal policy.

Disadvantages

- Discrete Action Space:** Limited to 15 fixed price levels, preventing fine-grained pricing decisions.
- Coarse State Aggregation:** Different surplus/demand quantities map to the same bucket, treating distinct market conditions as identical.
- Poor Scalability:** Suffers from the "curse of dimensionality" as agents, time periods, or variables grow.

Experiments with Neural Networks

State Space

Dimension	Seller	Buyer
s_1	hour / 47	hour / 47
s_2	surplus / 10	deficit / 10
s_3	best_bid / 0.5	best_bid / 0.5
s_4	best_ask / 0.5	best_ask / 0.5
s_5	avg_price / 0.5	avg_price / 0.5

Policy Net Layers

Layer	Dimensions	Parameters
Input	5 + 16 = 21	—
Linear + ReLU	21 → 64	1,408
Linear + ReLU	64 → 64	4,160
Linear + Sigmoid	64 → 1	65
Total (+ encoder)	—	≈ 5,857

Two Networks (seller + buyer) ≈ 11,700 per agent

Action Space: Continuous Pricing

The policy outputs a continuous pricing coefficient: $a \in [0, 1]$

The final trading price is computed as: $price = FiT + a \times (ToU - FiT)$

Exploration During Training

Gaussian exploration noise is added: $a_{raw} = a + N(0, \sigma)$

The resulting price remains bounded within: $[\$0.06, \$0.28]$

providing continuous pricing without discretization.

Reward Functions

Seller Reward

$$r_{seller} = q_{P2P} \times p_{P2P} + q_{grid} \times FiT + \beta(q_{P2P} \times p_{P2P} - q_{P2P} \times FiT)$$

- First term: revenue from P2P trading
- Second term: revenue from exporting excess energy to the grid
- Third term: pricing premium encouraging profitable P2P transactions

Buyer Reward

$$r_{buyer} = -(q_{P2P} \times p_{P2P} + q_{grid} \times ToU)$$

The buyer seeks to minimize total energy procurement costs.

REINFORCE Update Rule

The discounted return for each timestep is: $G_t = \sum_{k=t}^T \gamma^{k-t} r_k$

γ is the discount factor

r_k is the reward at step k

The policy objective is: $L(\theta) = -E[G_t \times \log_{\pi_{\theta}}(a_t | s_t, hist_t)]$

Parameters are updated using gradient descent:

$$\theta \leftarrow \theta - \eta \nabla_{\theta} L(\theta)$$

PolicyNet + TimeSeriesEncoder

- Last 12 market prices → TimeSeriesEncoder → 16-dim embedding
- Current state → 5-dim observation
- Concatenated input (21-dim) → 64 → 64 → 1 → Sigmoid
- Output: continuous action $a \in [0, 1]$

Two separate networks are used for buyers and sellers, sharing the same architecture but with independent weights.

TimeSeriesEncoder

The last 12 clearing prices are encoded as: $h_i = W_{price} p_i + PosEmb(i)$

The final representation is obtained by averaging all encoded vectors: $h_{enc} = \frac{1}{K} \sum_{i=0}^{K-1} h_i$

- Price projection: Linear(1 → 16)
- Positional embedding: Embedding(12 × 16)
- Total parameters: 224

Advantages

- Continuous Pricing:** Continuous price selection between FiT and ToU.
- Temporal Awareness:** Uses historical prices to capture market patterns.
- Scalability:** Generalizes to unseen states without large Q-tables.

Limitations

- Slower Convergence:** Requires more episodes than Q-Learning.
- Limited Interpretability:** Decisions are harder to explain than tabular methods.
- Higher Variance:** Training can be noisy and less stable.

Conclusion / Discussion

- P2P markets work. Autonomous learning agents increase peer-to-peer trading volume by +220% (76 → 245 kWh), reducing grid dependency for both buyers and sellers.
- Mutual gains are observed in the simulation. Seller revenue rises +10.6% while buyer costs fall -11.2% simultaneously, suggesting that learning-based pricing can improve outcomes for both sides under the modeled conditions.
- Neural networks find better equilibria. REINFORCE-NN achieves the lowest buyer cost (\$270.20) and highest P2P ratio (18.8%) through continuous pricing outcomes unreachable with Q-Learning's 15 discrete price levels.

- The model is market-agnostic. Although designed for energy trading, the agent-based auction framework applies to any decentralized market with buyers, sellers, and an outside option.

3-Way Comparison: Baseline | QL (Buyer+Seller) | NN (Buyer+Seller)

Metric	Baseline	QL (B+S)	NN (B+S)	ΔQL	ΔNN
P2P Ratio (%)	5.2	18.6	18.8	25.7%	26.2%
P2P Trading (kWh)	76.61	242.6	245.49	217%	220%
Grid Purchased / ToU (kWh)	1180.3	1014.4	1011.5	-14.1%	-14.3%
Grid Sold / FiT (kWh)	215	49.01	46.12	-77.2%	-78.5%
Avg. P2P Price (\$/kWh)	\$0.21	\$0.13	\$0.12	-34.8%	-42.6%

Seller Revenue & Buyer Cost Analysis

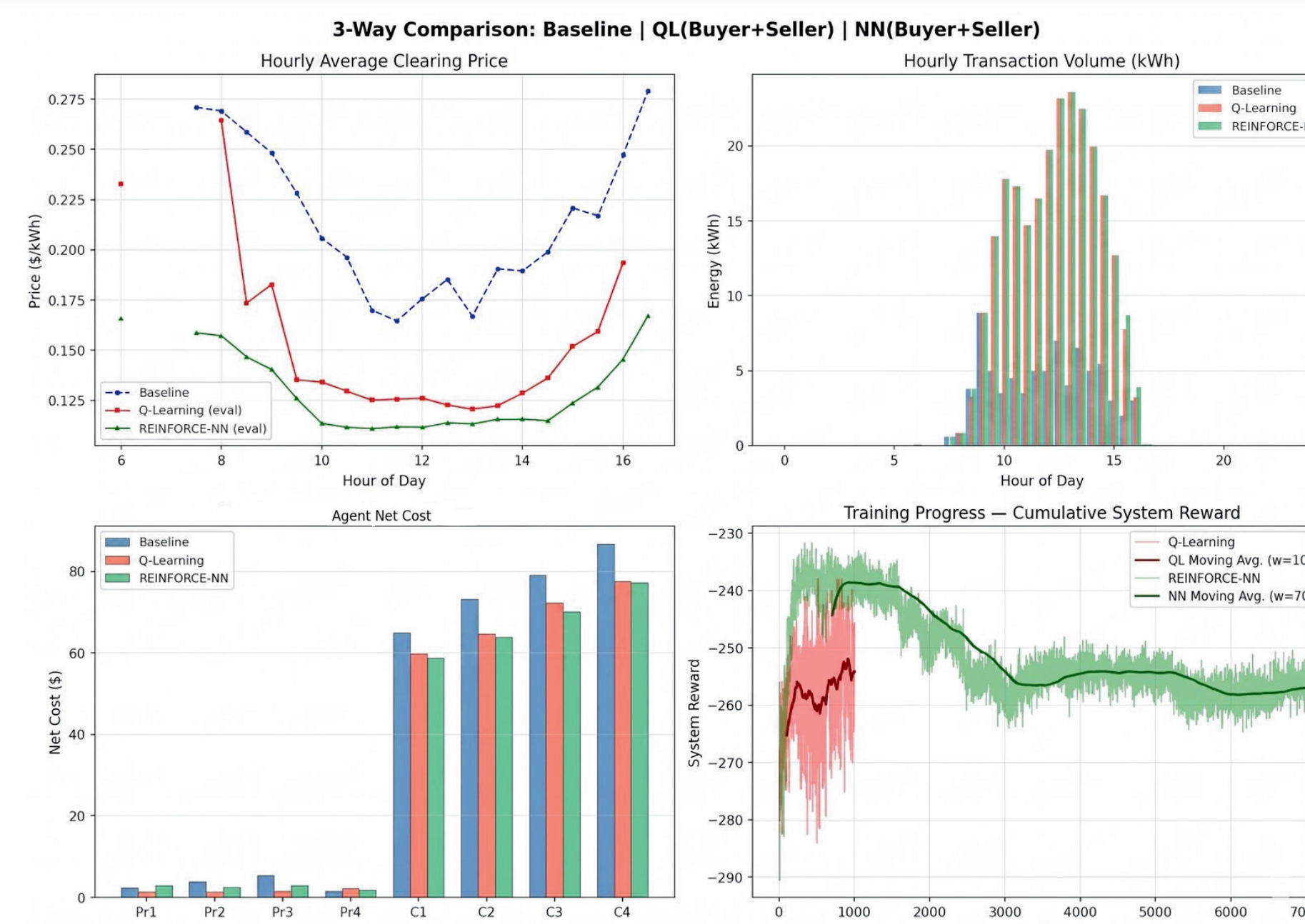
P2P is zero-sum: seller P2P income = buyer P2P cost

Seller Revenue

Metric	Baseline	QL (B+S)	NN (B+S)	ΔQL	ΔNN
P2P Income (\$)	15.7	32.39	28.86	106%	84%
Grid FiT Income (\$)	12.9	2.94	2.77	-77%	-79%
Total Seller Revenue (\$)	28.6	35.33	31.62	0.235	0.106

Buyer Costs

Metric	Baseline	QL (B+S)	NN (B+S)	ΔQL	ΔNN
P2P Purchase Cost (\$)	15.7	32.17	28.73	105%	83%
Grid ToU Cost (\$)	288.54	242.3	241.46	-16.0%	-16.3%
Total Buyer Cost (\$)	304.24	274.47	270.2	-9.8%	-11.2%



Technologies Used



References

- Taylor, J. A., and Mathieu, J. L., "Strategic Bidding in Electricity Markets with Only Renewables," ACC, 2016.
- Jiang, J., Li, Y., Hou, L., Ghahouri, M., Zhang, P., Yan, J., and Liu, Y., "Deep Reinforcement Learning-Based Bidding Strategies for Prosumers Trading in Double Auction-Based Transactive Energy Market," arXiv:2502.15774, 2025.
- Qiu, D., Wang, J., Wang, J., and Sirbac, G., "Multi-Agent Reinforcement Learning for Automated Peer-to-Peer Energy Trading in Double-Side Auction Market," IJCAI, 2021.
- E. L. Raman, S. R. Vetter, C. M. Kellett, and A. T. Murray, "Residential load and rooftop PV generation: an Australian distribution network dataset," International Journal of Sustainable Energy, vol. 36, no. 8, pp. 787-806, 2017, doi: 10.1080/14786451.2015.1100196.